

EcoCare Rover: A Low-Cost IoT-Enabled Indoor Healthcare Assistance Rover with Remote Monitoring and Zone-Based Delivery

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Abstract—The increasing demand for efficient healthcare support has encouraged the development of autonomous robotic systems capable of assisting with routine indoor service tasks. This paper presents EcoCare Rover, an Internet of Things (IoT)-enabled indoor healthcare delivery rover designed to support indoor healthcare delivery assistance through automated delivery, environmental monitoring, remote control, and cloud-based supervision. The proposed system is developed around an ESP32 microcontroller and integrates an HC-SR04 ultrasonic sensor for real-time obstacle detection, a DHT22 sensor for temperature and humidity monitoring, an L298N motor driver for locomotion control, and a buzzer-based alert mechanism for improving operational safety. The rover supports two operating modes: manual control through the Blynk mobile application and zone-based delivery using predefined motion sequences to selected locations, including bed, medicine desk, and water station zones. During operation, the system continuously processes sensor information, user commands, and rover status data to perform navigation and monitoring tasks. Operational parameters, including temperature, humidity, obstacle status, delivery progress, and selected destination, are synchronized with the ThingSpeak cloud platform to enable real-time visualization and historical data analysis. Experimental evaluation demonstrates the operational capability of manual and autonomous navigation, obstacle detection, environmental monitoring, and IoT-based remote supervision under controlled indoor conditions. The developed EcoCare Rover demonstrates the feasibility of a low-cost approach for indoor assistive delivery applications in healthcare environments.

Index Terms—Autonomous Delivery Rover, ESP32, IoT, Indoor Healthcare Robot, Blynk, ThingSpeak, Smart Healthcare, Environmental Monitoring, Mobile Robot

I. INTRODUCTION

Healthcare facilities are increasingly exploring intelligent robotic systems to reduce caregiver workload, improve operational efficiency, and support safer indoor logistics. Routine tasks such as transporting medicines, laboratory samples, medical supplies, water, and patient-support materials require considerable staff involvement and may reduce the time available for direct patient care. Recent studies indicate that autonomous mobile robots (AMRs), nursing robots, and assistive service robots are becoming valuable solutions for hospital logistics, elderly care, patient support, and remote healthcare services [1]–[4]. In addition, healthcare robots can

assist with transportation, monitoring, and routine support activities, reducing repetitive workload and improving service efficiency [5]–[8].

Despite these advancements, many existing healthcare robotic platforms require expensive hardware, Robot Operating System (ROS)-based frameworks, complex localization techniques, or high-cost sensing technologies, which can restrict their adoption in small hospitals, clinics, and healthcare environments with limited resources [9]. Recent low-cost nursing robots based on embedded platforms such as ESP32 have demonstrated the potential for affordable healthcare assistance; however, many systems still involve additional software complexity or lack integrated cloud-based environmental monitoring [10]. IoT-enabled healthcare robotics provides an effective approach by combining embedded control, wireless communication, mobile interfaces, and cloud-based visualization for real-time supervision [2], [3].

To address this requirement, this paper presents **EcoCare Rover**, an ESP32-based IoT-enabled autonomous indoor delivery rover designed for healthcare support. The proposed system integrates an HC-SR04 ultrasonic sensor for obstacle detection, a DHT22 sensor for environmental monitoring, an L298N motor driver for locomotion control, and a buzzer-based safety alert mechanism. The rover supports both manual operation through the Blynk mobile application and autonomous delivery to predefined zones, including bed, medicine station, and water station locations. Furthermore, operational parameters such as temperature, humidity, obstacle status, delivery status, and selected destination are uploaded to the ThingSpeak cloud platform for real-time visualization and historical monitoring.

The main contributions of this paper are summarized as follows:

- Development of a low-cost ESP32-based indoor healthcare delivery rover integrating sensing, control, and IoT communication.
- Implementation of dual operating modes through Blynk-based manual control and predefined zone-based delivery.
- Integration of obstacle detection, environmental monitoring, and IoT cloud synchronization within a lightweight

embedded healthcare delivery platform.

The remainder of this paper is organized as follows. Section II presents the related work on healthcare robotics. Section III introduces the proposed EcoCare Rover system. Section IV describes the overall architecture. Section V discusses the hardware components, followed by the software workflow in Section VI. Section VII presents the IoT integration. Section VIII describes the experimental setup, while Section IX discusses the results and performance evaluation. Finally, Section X concludes the paper.

II. RELATED WORK

Recent research has explored healthcare robots, autonomous mobile robots (AMRs), and IoT-based robotic platforms for hospital logistics and patient assistance applications [3], [11].

Suharjono *et al.* [10] developed a low-cost nursing robot using an ESP32 controller and Robot Operating System (ROS) for telemedicine applications. Their system demonstrated reliable remote communication and autonomous navigation. However, the dependency on a ROS-based framework increases software complexity and computational requirements, which may affect deployment in resource-constrained healthcare environments.

Wang *et al.* [11] reviewed the development of autonomous mobile robots in healthcare applications, including hospital logistics, elderly care, rehabilitation, and human-robot collaboration. The study identified the growing importance of intelligent delivery robots while also highlighting challenges related to hardware cost and advanced localization requirements.

Recent studies have also explored compact mobile robots specifically for medicine distribution inside hospitals. Such systems demonstrate the potential of autonomous robots for improving healthcare logistics; however, many implementations still rely on advanced navigation methods and do not focus on low-cost embedded platforms [12].

Schmidt *et al.* [9] analyzed mobile robotic assistance systems in hospitals and discussed challenges related to integration, safety, and deployment. However, their work mainly focused on hospital-level robotic assistance rather than low-cost standalone healthcare delivery platforms.

Unlike the reviewed systems, the proposed **EcoCare Rover** combines manual smartphone control, predefined zone-based delivery, obstacle detection, environmental monitoring, and cloud supervision in a single ESP32-based platform. The proposed design focuses on low-cost implementation and simple deployment by avoiding complex infrastructure requirements, making it a potential solution for small healthcare facilities and indoor assistance applications.

III. PROPOSED SYSTEM

The proposed **EcoCare Rover** is an ESP32-based IoT-enabled autonomous indoor delivery rover developed to support indoor healthcare assistance scenarios such as hospitals, elderly care centers, and smart indoor environments. The system combines predefined zone-based delivery, manual remote control, environmental monitoring, obstacle detection,

and cloud-assisted supervision within a compact and low-cost embedded platform.

The overall system architecture consists of sensing, control and decision, and IoT cloud layers, as illustrated in Fig. 1.

The rover operates in two modes: manual and autonomous. In manual mode, users control rover movement through the Blynk mobile application. In autonomous mode, users select predefined delivery destinations, and the rover executes the corresponding programmed navigation sequence while continuously monitoring obstacles for safe movement. The L298N motor driver controls the DC motors, and the buzzer provides an alert whenever an obstacle is detected.

For remote monitoring, the system integrates the Blynk and ThingSpeak platforms. Blynk enables real-time user interaction by providing manual control, operating mode selection, destination selection, and status monitoring. Meanwhile, ThingSpeak stores and visualizes environmental and operational parameters, including temperature, humidity, obstacle status, delivery status, and selected delivery zone [13], [14].

IV. OVERALL ARCHITECTURE

As shown in Fig. 1, the proposed EcoCare Rover follows a three-layer architecture consisting of the sensing layer, control and decision layer, and IoT cloud and user interface layer.

The sensing layer is responsible for collecting environmental and navigation information using a DHT22 sensor for temperature and humidity monitoring and an HC-SR04 ultrasonic sensor for obstacle detection. The acquired sensor data are processed by the ESP32 controller, enabling real-time decision making and rover control.

The control and decision layer serves as the main processing unit of the system. The ESP32 integrates sensor processing, navigation control, wireless communication, and delivery management within a single embedded platform. Based on user commands and sensor information, the rover operates in either manual or autonomous mode. In manual mode, movement commands are received through the Blynk application, whereas autonomous mode performs predefined zone-based delivery according to the selected destination. The L298N motor driver controls the DC motors, while the buzzer generates obstacle alerts when required. Furthermore, the integrated Wi-Fi capability of the ESP32 enables direct communication with IoT platforms without requiring additional communication hardware [15].

The IoT cloud and user interface layer enables remote monitoring and control through the Blynk and ThingSpeak platforms. The Blynk application provides user interaction for manual navigation, operating mode selection, delivery zone selection, and real-time status monitoring. Meanwhile, ThingSpeak stores and visualizes environmental and operational parameters, including temperature, humidity, obstacle status, delivery status, and selected delivery zone [16].

This layered design integrates sensing, control, robotic actuation, and IoT communication within a modular framework.

TABLE I
COMPARISON OF SELECTED HEALTHCARE ROBOTICS STUDIES AND THE PROPOSED ECOCare ROVER

Study	Type	ESP32	IoT	Manual	Auto.	Obstacle	Env.	Cloud	Low Cost	Healthcare
Suharjono et al. (2025) [10]	Prototype	✓	✓	✓	✓	✓	×	✓	×	✓
Shin et al. (2025) [3]	Review	×	Partial	×	N/A	Partial	Partial	Partial	N/A	✓
Wang et al. (2024) [11]	Survey	×	Partial	N/A	N/A	Partial	Partial	Partial	N/A	✓
Silvera-Tawil (2024) [1]	Review	×	Partial	N/A	N/A	Partial	Partial	Partial	N/A	✓
Shaw et al. (2025) [6]	Review	×	Partial	×	Partial	×	Partial	Partial	N/A	✓
Soriano et al. (2022) [5]	Review	×	×	×	N/A	×	Partial	×	N/A	✓
Adeyemo et al. (2025) [7]	Review	×	Partial	×	N/A	×	Partial	Partial	N/A	✓
Proposed EcoCare Rover	Prototype	✓	✓	✓	✓	✓	✓	✓	✓	✓

Note: ✓ indicates available features, × indicates unavailable features, Partial indicates partially supported or discussed features, and N/A denotes not applicable.

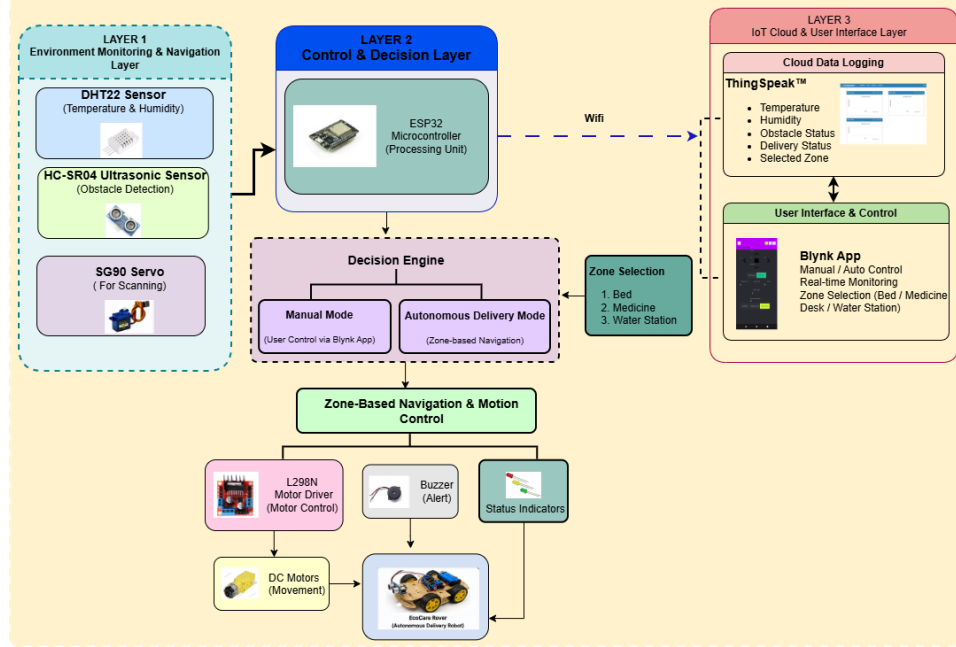


Fig. 1. System architecture of the proposed EcoCare Rover illustrating sensing, decision-making, IoT communication, and robotic actuation layers.

V. HARDWARE COMPONENTS

The proposed EcoCare Rover is developed using low-cost and commercially available hardware components to support indoor delivery, environmental monitoring, obstacle detection, and IoT-based supervision. The hardware design prioritizes affordability, modularity, and ease of implementation. The main hardware components used in the proposed system are summarized in Table II.

The ESP32 development board acts as the primary controller of the system, handling sensor data acquisition, navigation control, motor operation, and IoT communication. Its built-in Wi-Fi capability removes the need for an external communication module, which helps reduce hardware complexity and overall system cost [15].

The HC-SR04 ultrasonic sensor is employed for obstacle detection during rover movement, while the DHT22 sensor is used to monitor temperature and humidity conditions. The L298N motor driver manages the DC geared motors for

TABLE II
HARDWARE COMPONENTS OF THE PROPOSED ECOCare ROVER

Component	Function
ESP32 Development Board	Main Controller and Wi-Fi Communication
HC-SR04 Ultrasonic Sensor	Obstacle Detection
DHT22 Sensor	Temperature and Humidity Monitoring
SG90 Servo Motor	Sensor Mount Adjustment
L298N Motor Driver	DC Motor Control
DC Gear Motors	Rover Locomotion
Buzzer	Obstacle Alert Notification
Li-ion Battery Pack	Power Supply
Robot Chassis	Mechanical Platform

locomotion, and the buzzer provides an audible alert when an obstacle is detected.

The estimated cost represents approximate component prices collected from the local Bangladeshi market and may vary depending on supplier, component availability, and exchange rate fluctuations.

TABLE III
ESTIMATED PROTOTYPE HARDWARE COST OF THE PROPOSED ECO-CARE ROVER IN BANGLADESH

Component	Estimated Cost (BDT)
ESP32 Development Board	850
HC-SR04 Ultrasonic Sensor	200
DHT22 Sensor	550
SG90 Servo Motor	300
L298N Motor Driver	400
DC Gear Motors	850
Robot Chassis	1100
Battery Pack	900
Miscellaneous Components	500
Total Estimated Cost	5650

VI. SOFTWARE WORKFLOW

The software workflow of the proposed EcoCare Rover is developed to coordinate sensor processing, navigation control, and IoT-based monitoring through the ESP32 embedded platform. The complete operational sequence is illustrated in Fig. 2. After system initialization, the ESP32 configures the connected sensors, motor driver, Wi-Fi communication, Blynk service, and ThingSpeak cloud connection. During operation, the controller continuously processes sensor measurements, user commands, and rover status information to perform the required tasks.

The developed software supports two operating modes: manual and autonomous. In manual mode, movement commands received from the Blynk mobile application are processed by the ESP32 to control the rover movement. In autonomous mode, the user selects a required delivery zone, and the controller follows the corresponding predefined motion sequence while continuously checking obstacles using the ultrasonic sensor. The rover does not rely on GPS, wheel encoders, line-following sensors, or localization algorithms to determine its position. Instead, each delivery zone is assigned a fixed movement pattern based on forward motion, turning direction, and delay time. After completing the assigned sequence, the rover stops and considers the selected destination as reached before updating the delivery status. Sensor readings, including temperature, humidity, obstacle condition, and delivery status, are periodically transmitted to the ThingSpeak platform, while the Blynk application is used for real-time monitoring and user control. This approach provides a simple and low-cost control strategy suitable for the proposed indoor healthcare assistance application [16], [17].

VII. IOT INTEGRATION

The IoT framework of the proposed EcoCare Rover provides wireless communication between the rover, mobile interface, and cloud platform. The ESP32 microcontroller acts as the primary IoT gateway, providing wireless communication between the rover, Blynk mobile application, and ThingSpeak cloud platform. The integration of cloud connectivity enables remote monitoring and storage of operational data for the proposed robotic system [13]. The Blynk application provides the human-machine interface for manual and autonomous

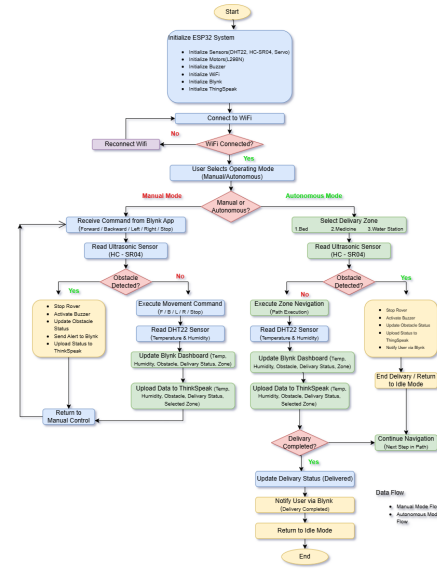


Fig. 2. Operational workflow of EcoCare Rover showing initialization, dual-mode control, obstacle-aware navigation, and cloud synchronization.

mode selection, rover control, delivery zone selection, and real-time monitoring of temperature, humidity, obstacle status, and delivery status [17].

For cloud-based storage and visualization, the ESP32 periodically uploads operational data to the ThingSpeak platform, including temperature, humidity, obstacle status, delivery status, and selected delivery zone. ThingSpeak provides graphical visualization and historical data analysis for monitoring rover performance [16]. The communication process involves local sensor processing by the ESP32, data transmission to Blynk for real-time interaction, and synchronization with ThingSpeak for cloud storage. This bidirectional communication enables responsive remote control and continuous monitoring through the IoT platform.

VIII. EXPERIMENTAL SETUP

The proposed EcoCare Rover was experimentally evaluated in a controlled indoor environment to assess its navigation performance, obstacle detection capability, environmental monitoring functionality, and IoT-based remote supervision. The experimental setup consisted of predefined healthcare delivery locations, including a patient bed, medicine station, and water station, to create a controlled indoor testing environment. The test area included static obstacles placed along the navigation path to evaluate obstacle detection performance.

The developed prototype shown in Fig. 5 integrates an ESP32 controller, HC-SR04 ultrasonic sensor, DHT22 environmental sensor, DC motor-based locomotion system, and a healthcare delivery platform. The compact structure demonstrates the feasibility of implementing a low-cost IoT-enabled indoor delivery rover using readily available hardware components.

The embedded software was developed using Arduino IDE with ESP32 support. The Blynk platform was utilized for real-

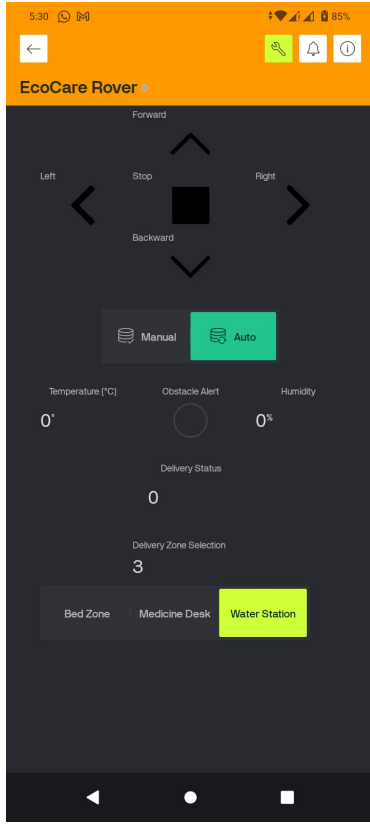


Fig. 3. Blynk-based user interface for rover control, destination selection, and real-time system monitoring.



Fig. 4. ThingSpeak cloud dashboard showing real-time environmental monitoring and operational status visualization of the EcoCare Rover.

time mobile interaction, while ThingSpeak was used for cloud-based data logging and visualization. The rover was evaluated under two operating modes: manual control through the Blynk application and predefined zone-based delivery to predefined destinations.

The experimental evaluation included manual navigation response, autonomous delivery performance, obstacle detec-



Fig. 5. Developed EcoCare Rover prototype integrating the ESP32 controller, ultrasonic obstacle detection module, motor drive system, and healthcare delivery platform.

TABLE IV
EXPERIMENTAL CONFIGURATION

Parameter	Specification
Programming IDE	Arduino IDE 2.x
Programming Language	C++
Cloud Platform	ThingSpeak
Mobile Platform	Blynk IoT
Communication	Wi-Fi (802.11 b/g/n)
Power Supply	7.4 V Li-ion Battery Pack
Obstacle Threshold	20 cm
ThingSpeak Update Interval	15 s
Sensor Sampling Interval	2 s
Operating Modes	Manual and Autonomous
Delivery Zones	Bed, Medicine, Water Station

tion, environmental monitoring, and cloud synchronization tests. The autonomous delivery evaluation was repeated for 30 trials across predefined delivery zones, while obstacle detection performance was evaluated using 50 obstacle detection tests at different positions along the navigation path. During autonomous operation, the rover selected predefined delivery zones and executed the corresponding navigation sequence while continuously monitoring obstacles through the ultrasonic sensor.

The hardware and software configuration used during the experiments is summarized in Table IV.

IX. RESULTS AND DISCUSSION

The proposed EcoCare Rover was experimentally evaluated to assess its autonomous delivery performance, obstacle detection capability, environmental monitoring, and IoT-based remote supervision. The experiments verified the operation of the sensing, communication, cloud monitoring, and robotic control modules.

TABLE V
PERFORMANCE EVALUATION OF THE PROPOSED ECO CARE ROVER

Performance Metric	Measured Result
Autonomous Delivery Success Rate	96.7% (30 trials)
Obstacle Detection Performance	98.2% (50 tests)
Average Delivery Time	17.6 s
ThingSpeak Data Upload Success	100%

During manual operation, movement commands transmitted through the Blynk mobile application were successfully received by the ESP32 controller, enabling responsive control of forward, backward, left, right, and stop movements. In autonomous mode, the rover successfully performed predefined zone-based deliveries to selected destinations, including bed, medicine station, and water station. The ultrasonic sensing mechanism provided obstacle awareness by stopping the rover and activating the buzzer whenever an obstacle was detected within the defined safety threshold.

The DHT22 sensor continuously monitored temperature and humidity conditions, while the collected data were displayed through the Blynk dashboard and uploaded to the ThingSpeak cloud platform. Figure 4 illustrates the cloud visualization of environmental parameters and rover operational status. The Blynk interface enabled remote control, operating mode selection, delivery zone selection, and real-time monitoring of rover conditions, as shown in Fig. 3.

Table V summarizes the overall performance evaluation of the proposed system. The performance metrics were obtained from repeated experimental trials conducted under controlled indoor conditions, and the reported values represent the average performance observed during testing. The obtained results demonstrate reliable navigation performance, effective obstacle detection, stable environmental monitoring, and successful IoT-based cloud synchronization under controlled indoor conditions. The response time of the rover was not considered as a separate performance metric because the current prototype does not include a dedicated latency measurement mechanism.

The delivery success rate was calculated based on the ratio of successfully completed delivery trials to the total number of autonomous delivery trials. Similarly, Obstacle Detection Performance was calculated from the number of correctly detected obstacle events compared with the total obstacle test cases. These measurements were obtained from repeated experiments conducted under controlled indoor conditions.

A small number of failures were observed during testing, mainly due to the limitations of predefined motion control and ultrasonic sensing.

X. CONCLUSION

This paper presented **EcoCare Rover**, an ESP32-based IoT-enabled autonomous indoor healthcare delivery rover designed for healthcare assistance. The proposed system integrates zone-based autonomous delivery, manual Blynk-based control, environmental monitoring, obstacle detection, and ThingSpeak cloud-based supervision within a low-cost embedded platform.

Experimental results demonstrated successful manual and autonomous navigation, real-time obstacle detection, environmental monitoring, and cloud synchronization. The integration of embedded sensing, wireless communication, and IoT services provides a practical approach for indoor healthcare delivery support.

Although the proposed system achieved reliable performance in controlled environments, the current navigation approach is limited by predefined routes and ultrasonic-based obstacle detection. Future work will focus on improving navigation autonomy through advanced localization, SLAM, adaptive path planning, vision-based perception, and intelligent decision-making techniques.

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